Lab Assignment 4

In [1]:
```
import pandas as pd
import numpy as np
df = pd.read_csv("BostonHousing.csv")
```

In [2]:
```
df
```

Out[2]:

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	b	l
0	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296	15.3	396.90	4
1	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8	396.90	9
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8	392.83	4
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	394.63	2
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	396.90	5
...	...	...	...	...	...	...	...	...	...	...	...	...	
501	0.06263	0.0	11.93	0	0.573	6.593	69.1	2.4786	1	273	21.0	391.99	9
502	0.04527	0.0	11.93	0	0.573	6.120	76.7	2.2875	1	273	21.0	396.90	9
503	0.06076	0.0	11.93	0	0.573	6.976	91.0	2.1675	1	273	21.0	396.90	5
504	0.10959	0.0	11.93	0	0.573	6.794	89.3	2.3889	1	273	21.0	393.45	6
505	0.04741	0.0	11.93	0	0.573	6.030	80.8	2.5050	1	273	21.0	396.90	7

506 rows × 14 columns

In [4]:
```
df.isnull().sum()
```

Out[4]:
```
crim       0
zn         0
indus      0
chas       0
nox        0
rm         5
age        0
dis        0
rad        0
tax        0
ptratio    0
b          0
lstat      0
medv       0
dtype: int64
```

In [5]:
```
df["rm"] = df["rm"].fillna(df["rm"].mean())
```

In [6]:
```
df.isnull().sum()
```

Out[6]:
```
crim       0
zn         0
indus      0
chas       0
nox        0
rm         0
age        0
dis        0
rad        0
tax        0
ptratio    0
b          0
lstat      0
medv       0
dtype: int64
```

In [7]:
```
y=df["medv"]
x=df.drop("medv",axis=1)
```

In [8]:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
```

In [10]:
```
x_train, x_test, y_train, y_test = train_test_split(x,y,test_size = 0.2)
```

In [11]:
```
lm = LinearRegression()
model=lm.fit(x_train, y_train)
```

In [12]:
```
y_train_pred = lm.predict(x_train)
y_test_pred = lm.predict(x_test)
```

In [15]:
```
from sklearn.metrics import mean_squared_error, r2_score, mean_absolute_error
```

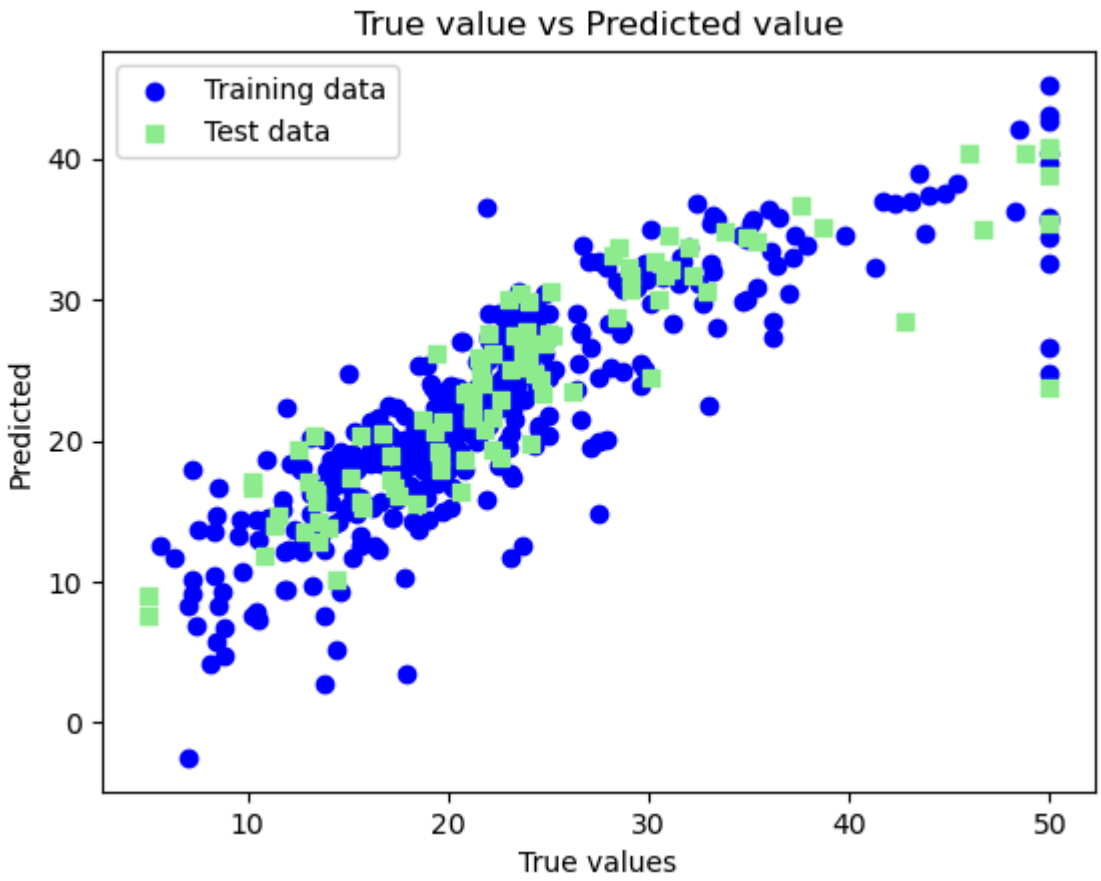
In [16]:
```
mse = mean_squared_error(y_test, y_test_pred)
print(mse)
mae = mean_absolute_error(y_test, y_test_pred)
print(mae)
rmse=np.sqrt(mse)
rmse
```

```
25.780286234077913
3.497218141904704
```

Out[16]:
```
np.float64(5.077429096903069)
```

In [17]:
```
from sklearn.metrics import r2_score
rscore = r2_score(y_test, y_test_pred)
rscore
```

Out[17]:
```
0.7464344533761373
```

In [20]:
```
import matplotlib.pyplot as plt
plt.scatter(y_train ,y_train_pred,c='blue',marker='o',label='Training data')
plt.scatter(y_test,y_test_pred ,c='lightgreen',marker='s',label='Test data')
plt.xlabel('True values')
plt.ylabel('Predicted')
plt.title("True value vs Predicted value")
plt.legend(loc= 'upper left')
#plt.hlines(y=0,xmin=0,xmax=50)
```

```
plt.plot()
plt.show()
```